

Exp no :02

Date :

Recover deleted or damaged files from a storage device using Test Disk

Description:

TestDisk step by step to recover a missing partition and repair a corrupted one.

Log creation



```
kali@kali: ~
Session Actions Edit View Help
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

TestDisk is free data recovery software designed to help recover lost
partitions and/or make non-booting disks bootable again when these symptoms
are caused by faulty software, certain types of viruses or human error.
It can also be used to repair some filesystem errors.

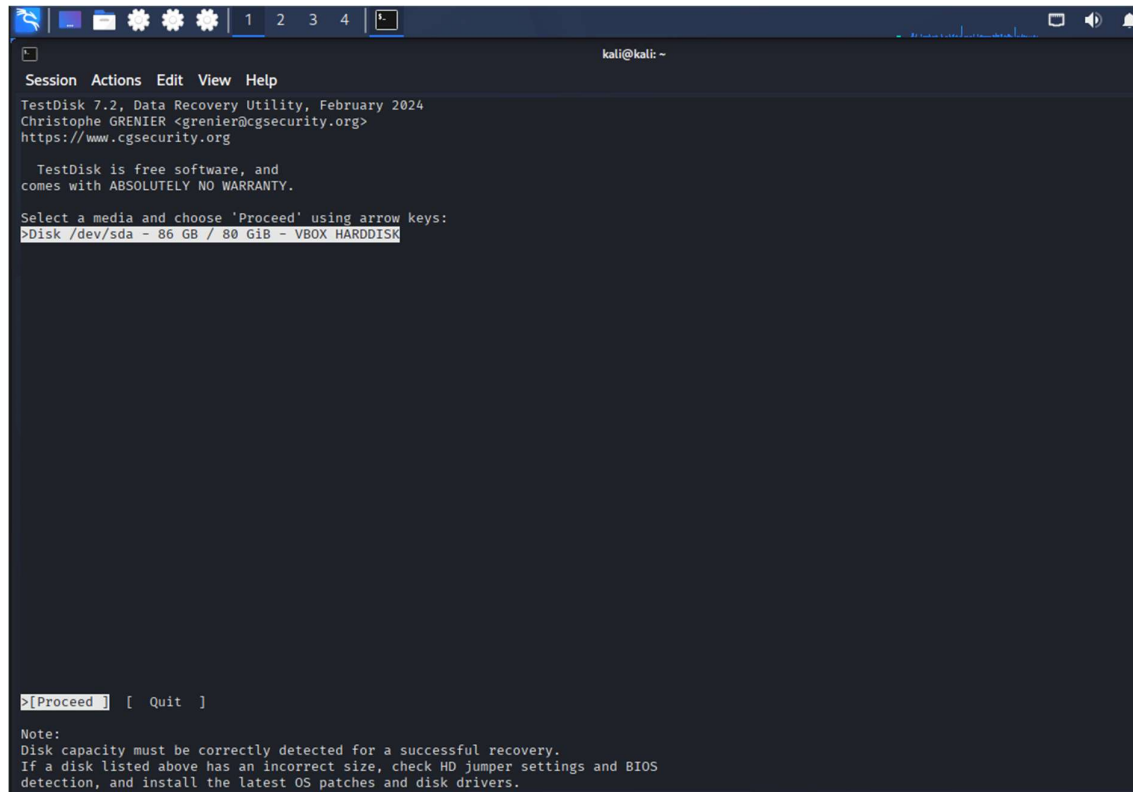
Information gathered during TestDisk use can be recorded for later
review. If you choose to create the text file, testdisk.log, it
will contain TestDisk options, technical information and various
outputs; including any folder/file names TestDisk was used to find and
list onscreen.

Use arrow keys to select, then press Enter key:
> [ Create ] Create a new log file
[ Append ] Append information to log file
[ No Log ] Don't record anything
```

- Choose Create to instruct Testdisk to create a [log file](#) containing technical information and messages, unless you have a reason to append data to the [log](#) or you execute TestDisk from read only media and must create the [log](#) elsewhere.
- Choose None if you do not want messages and details of the process to be written into a [log file](#) (useful if for example Testdisk was started from a read-only location).
- Press Enter to proceed.

Disk Selection

All hard drives should be detected and listed with the correct size by TestDisk:

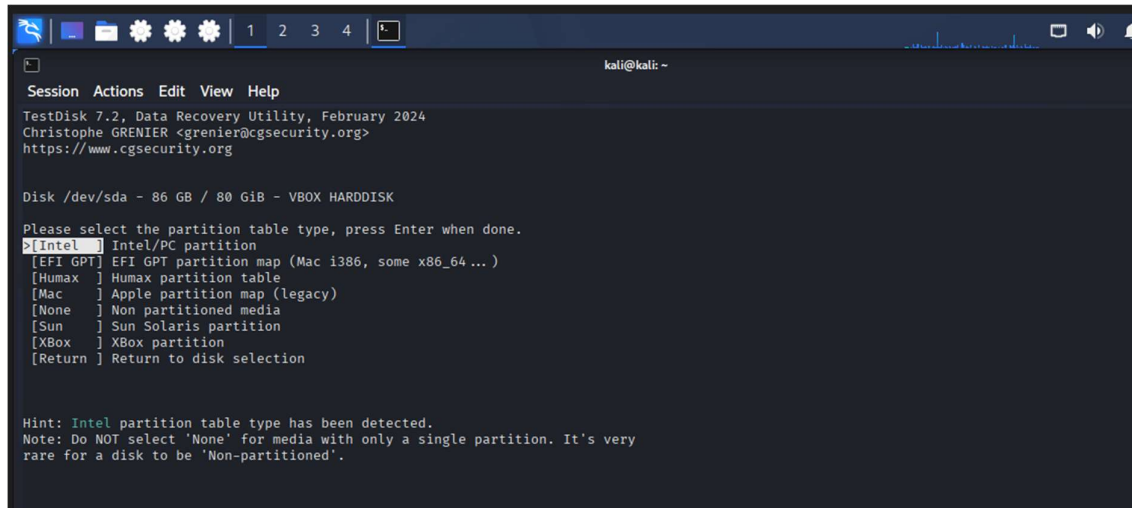


- Use up/down arrow keys to select your hard drive with the lost partition/s.
- Press Enter to Proceed.

If available, use raw device `/dev/sda*` for faster data transfer.

Partition table type selection

TestDisk displays the partition table types.



```
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Disk /dev/sda - 86 GB / 80 GiB - VBOX HARDDISK

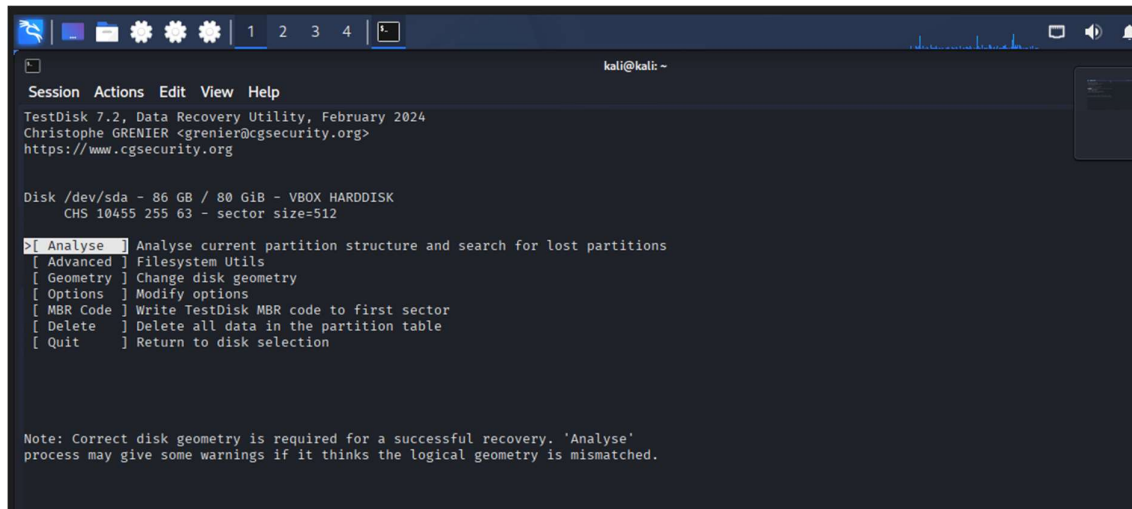
Please select the partition table type, press Enter when done.
>[Intel] Intel/PC partition
[EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
[Humax] Humax partition table
[Mac] Apple partition map (legacy)
[None] Non partitioned media
[Sun] Sun Solaris partition
[XBox] Xbox partition
[Return] Return to disk selection

Hint: Intel partition table type has been detected.
Note: Do NOT select 'None' for media with only a single partition. It's very
rare for a disk to be 'Non-partitioned'.
```

- Select the partition table type - usually the default value is the correct one as TestDisk auto-detects the partition table type.
- Press Enter to Proceed.

Current Partition Table Status

TestDisk displays the menus (also see [TestDisk Menu Items](#)).



```
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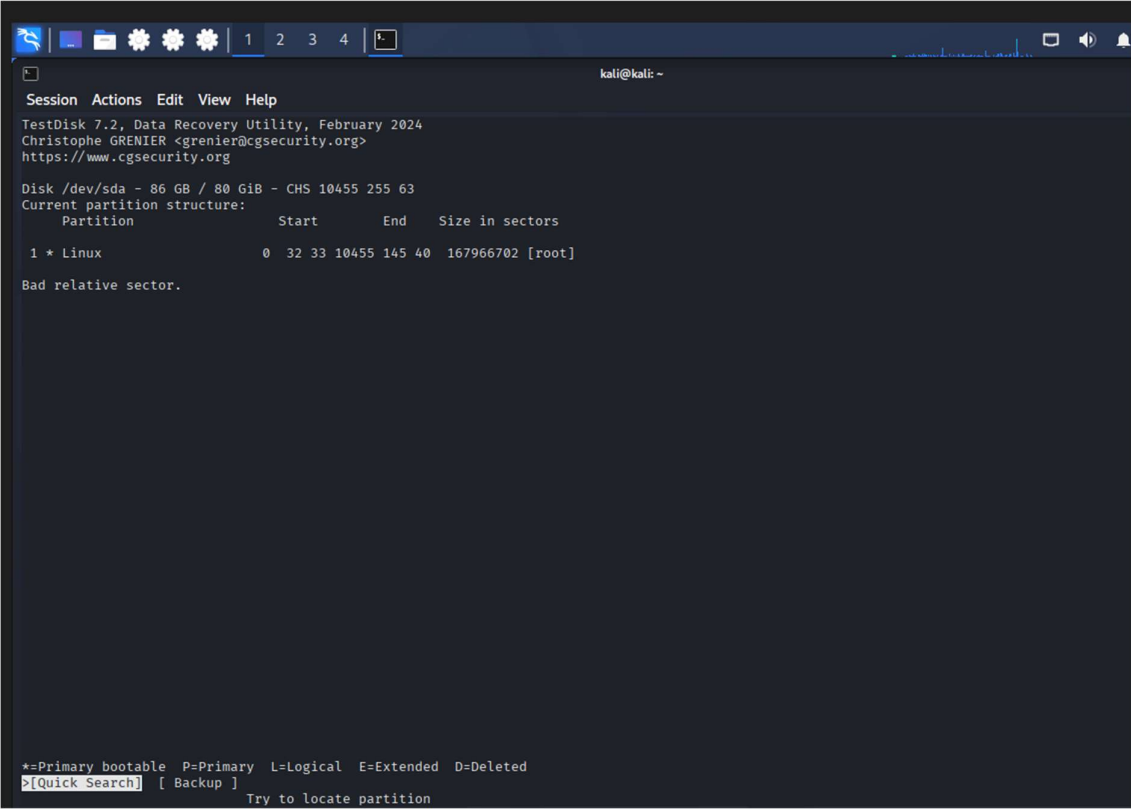
Disk /dev/sda - 86 GB / 80 GiB - VBOX HARDDISK
CHS 10455 255 63 - sector size=512

>[Analyse] Analyse current partition structure and search for lost partitions
[Advanced] Filesystem Utils
[Geometry] Change disk geometry
[Options] Modify options
[MBR Code] Write TestDisk MBR code to first sector
[Delete] Delete all data in the partition table
[Quit] Return to disk selection

Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatched.
```

- Use the default menu "Analyse" to check your current partition structure and search for lost partitions.
- Confirm at Analyse with Enter to proceed.

Now, your current partition structure is listed. Examine your current partition structure for missing partitions and errors.



```

kali@kali: ~
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Disk /dev/sda - 86 GB / 80 GiB - CHS 10455 255 63
Current partition structure:
  Partition      Start      End      Size in sectors
  1 * Linux      0  32 33 10455 145 40  167966702 [root]
Bad relative sector.

**Primary bootable P=Primary L=Logical E=Extended D=Deleted
>[Quick Search] [Backup] Try to locate partition

```

The first partition is listed twice which points to a corrupted partition or an invalid partition table entry.

Invalid NTFS boot points to a faulty NTFS boot sector, so it's a corrupted filesystem.

Only one logical partition (label Partition 2) is available in the extended partition. One logical partition is missing.

- Confirm at **Quick Search** to proceed.

Quick Search for Partitions

During the **Quick Search**, TestDisk has found two partitions including the missing logical partition labeled *Partition 3*.

- Highlight this partition and press **p** to list your files (to go back to the previous display, press q to Quit, Files listed in red are deleted entries).

All directories and data are correctly listed.

- Press Enter to proceed.

Save the Partition table or search for more partitions

```

kali@kali: ~
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Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk /dev/sda - 86 GB / 80 GiB - CHS 10455 255 63
Partition      Start      End      Size in sectors
>* Linux        0  32 33 10455 145 58  167966720 [root]

Structure: Ok. Use Up/Down Arrow keys to select partition.
Use Left/Right Arrow keys to CHANGE partition characteristics:
**=Primary bootable P=Primary L=Logical E=Extended D=Deleted
Keys A: add partition, L: load backup, T: change type, P: list files,
Enter: to continue
ext4 blocks=4096 Large_file Sparse_SB Recover, 85 GB / 80 GiB

```

- **When all partitions are available** and data correctly listed, you should go to the menu **Write** to save the partition structure. The menu Extd Part gives you the opportunity to decide if the extended partition will use all available disk space or only the required (minimal) space.
- **Since a partition, the first one, is still missing**, highlight the menu **Deeper Search** (if not done automatically already) and press Enter to proceed.


```

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1 * Linux 0 32 33 10455 145 58 167966720 [root]
Directory /

>drwxr-xr-x 0 0 4096 16-Jun-2026 10:56 .
drwxr-xr-x 0 0 4096 16-Jun-2026 10:56 ..
drwx----- 0 0 16384 16-Jun-2026 10:55 lost+found
drwxr-xr-x 0 0 4096 16-Jun-2026 10:29 srv
drwxr-xr-x 0 0 4096 16-Jun-2026 10:23 media
lrwxrwxrwx 0 0 7 16-Jun-2026 06:14 lib
drwxr-xr-x 0 0 4096 16-Jun-2026 10:23 run
lrwxrwxrwx 0 0 9 16-Jun-2026 06:14 lib64
drwxr-xr-x 0 0 4096 16-Jun-2026 10:23 mnt
lrwxrwxrwx 0 0 8 16-Jun-2026 06:14 sbin
drwx----- 0 0 4096 29-Jul-2026 01:29 root
drwxrwxrwt 0 0 4096 16-Jun-2026 10:23 tmp
drwxr-xr-x 0 0 4096 16-Jun-2026 10:27 opt
drwxr-xr-x 0 0 4096 16-Jun-2026 10:23 dev
drwxr-xr-x 0 0 4096 16-Jun-2026 06:14 sys
lrwxrwxrwx 0 0 9 16-Jun-2026 10:27 lib32
lrwxrwxrwx 0 0 7 16-Jun-2026 06:14 bin
drwxr-xr-x 0 0 4096 16-Jun-2026 10:30 home
drwxr-xr-x 0 0 4096 16-Jun-2026 10:56 boot
drwxr-xr-x 0 0 4096 16-Jun-2026 06:14 proc
drwxr-xr-x 0 0 12288 7-Sep-2026 06:17 etc
drwxr-xr-x 0 0 4096 16-Jun-2026 10:27 usr
drwxr-xr-x 0 0 4096 20-Jul-2026 03:08 var
-rw----- 0 0 1000000000 16-Jun-2026 10:56 swap
lrwxrwxrwx 0 0 34 16-Jun-2026 10:56 initrd.img.old
lrwxrwxrwx 0 0 31 16-Jun-2026 10:56 vmlinuz.old
lrwxrwxrwx 0 0 34 16-Jun-2026 10:56 initrd.img
lrwxrwxrwx 0 0 31 16-Jun-2026 10:56 vmlinuz

Next
Use Right to change directory, 'h' to hide deleted files
'q' to quit, 'i' to select the current file, 'a' to select all files
'c' to copy the selected files, 'C' to copy the current file

```

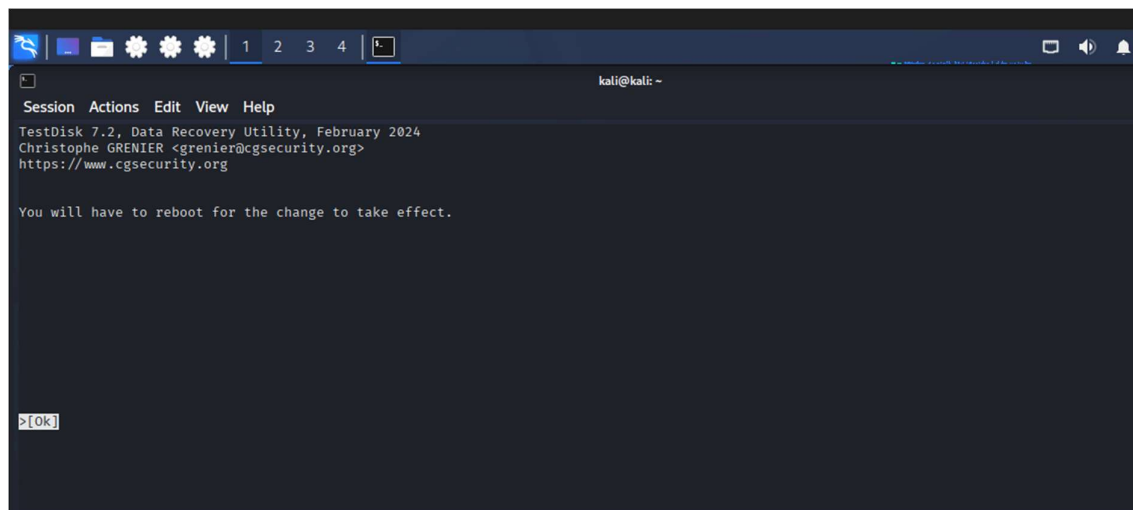
It works, your files are listed, you have found the correct partition!

- Use the left/right arrow to navigate into your folders and watch your files for more verification

Note: FAT directory listing is limited to 10 clusters - some files may not appear but it doesn't affect recovery.

- Press q for Quit to go back to the previous display.
- The available status are Primary, * bootable, Logical and Deleted.

Using the left/right arrow keys, change the status of the selected partition from Deleted to **Logical**. This way you will be able to recover this partition.



TestDisk displays *You have to restart your Computer to access your data* so press Enter a last time and reboot your computer.